2.3: The formation and function of molecules and ionic compounds depend on chemical bonding between atoms

Now that we have looked at the structure of atoms, we can move up the hierarchy of organization and see how atoms combine to form molecules and ionic compounds. Atoms with incomplete valence shells can interact with certain other atoms in such a way that each partner atom completes its valence shell: The atoms either share or transfer valence electrons. These interactions usually result in atoms staying close together, held by attractions called **chemical bonds**. The strongest kinds of chemical bonds are covalent bonds in molecules and ionic bonds in dry ionic compounds. (Ionic bonds in aqueous, or water-based, solutions are weak interactions, as we will see later.)

Animation

Introduction to Chemical Bonds

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Covalent Bonds

A **covalent bond** is the sharing of a pair of valence electrons by two atoms. For example, let's consider what happens when two hydrogen atoms approach each other. Recall that hydrogen has 1 valence electron in the first shell, but the shell's capacity is 2 electrons. When the two hydrogen atoms come close enough for their 1s orbitals to overlap, they can share their electrons **(Figure 2.9)**. Each hydrogen atom is now associated with 2 electrons in what amounts to a completed valence shell. Two or more atoms held together by covalent bonds constitute a **molecule**, in this case a hydrogen molecule.

Figure 2.9

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Formation of a covalent bond.
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Formation of a covalent bond.

Begin with 2 hydrogen atoms (2 H). First. In each hydrogen atom, the single electron is held in its orbital by its attraction to the proton in the nucleus. Second. When two hydrogen atoms approach each other, the electron of each atom is also attracted to the proton in the other nucleus. Third. The two electrons become shared in a covalent bond, forming an H 2 molecule.

Figure 2.10a shows several ways of representing a hydrogen molecule. Its *molecular formula*, H2, simply indicates that the molecule consists of two atoms of hydrogen. Electron sharing can be depicted by an electron distribution diagram or by a *Lewis dot structure*, in which element symbols are surrounded by dots that represent the valence electrons (H:H). We can also use a *structural formula*, H—H, where the line represents a **single bond**, a pair of shared electrons. A *space-filling model* comes closest to representing the actual shape of the molecule. (You may also be familiar with ball-and-stick models, which are shown in Figure 2.15.)

Figure 2.10

Covalent bonding in four molecules.

Table 2.10, covalent bonding in four molecules.

The number of electrons required to complete an atom's valence shell generally determines how many covalent bonds that atom will form. This figure shows several ways of indicating covalent bonds.

Name and Molecular Formula	Electron Distribution Diagram	Lewis Dot Structure and Structural Formula	Space Filling Model
Hydrogen (H 2). Two hydrogen atoms share one pair of electrons, forming a single bond.	Shell models of two hydrogen atoms, each sharing their lone electron between them to form a single covalent bond that fills the outer electron shell of both atoms.	Dot Formula: A pair of capital H letters with two dots between them, where the dots represent a single pair of shared electrons.	Two small circles pushed together represent two hydrogen atoms forming a single covalent bond.
Structural formula: Two capital H letters with a single line between them, where the single line represents a shared			

pair of electrons, and therefore a single covalent bond.			
Oxygen (O 2). Two oxygen atoms share two pairs of electrons, forming a double bond.	Shell models of two oxygen atoms with two electrons in the first shells, and six electrons in each second shell. Each oxygen atom shares two pairs of electrons in the outer shell, forming a double covalent bond.	Dot formula: Two capital O letters with two pairs of dots between them, representing a double covalent bond. A pair of dots is above and below each O, showing four unbounded electrons on each atom.	Two large circles pressed together, representing two bonded oxygen atoms.
Structural formula: Two Os with two lines between them, representing a double covalent bond.			
Water (H20). Two hydrogen atoms and one oxygen atom are joined by single bonds, forming a molecule of water.	Water molecule depicted as a shell model of an oxygen atom sharing a pair of electrons with each of two hydrogen atoms in forming covalent bonds. In addition to the shared electrons, oxygen has 2 electrons in the first shell.	Dot formula: O with two dots between it and each of two H atoms. The O additionally has two dots above and two on the side indicating unbounded electrons.	A large circle is an oxygen atom pressed to two smaller circles representing hydrogen atoms bonded to the O.
Structural formula: O with lines to two Hs.			
Methane (CH4). Four hydrogen atoms can satisfy the valence of one carbon atom, forming methane.	Central carbon sharing a pair of electrons with each of four hydrogen atoms, for a total of 8 electrons in carbon's outer shell and two electrons in the outer shell of each hydrogen. In addition to the shared	Dot formula: central C surrounded by 4 Hs, with a pair of dots between each H and the C.	Larger central circle representing carbon surrounded by four smaller circles representing hydrogens.

	electrons, carbon has 2 electrons in its first shell.		
Structural formula: A central C surrounded by 4 Hs, with lines between each H and the C.			

Animation

Covalent Bonds

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Oxygen has 6 electrons in its second electron shell and therefore needs 2 more electrons to complete its valence shell. Two oxygen atoms form a molecule by sharing *two* pairs of valence electrons **(Figure 2.10b)**. The atoms are thus joined by what is called a **double bond** (O═O).

Each atom that can share valence electrons has a bonding capacity corresponding to the number of covalent bonds the atom can form. When the bonds form, they give the atom a full complement of electrons in the valence shell. The bonding capacity of oxygen, for example, is 2. This bonding capacity is called the atom's **valence** and usually equals the number of electrons required to complete the atom's outermost (valence) shell. See if you can determine the valences of hydrogen, oxygen, nitrogen, and carbon by studying the electron distribution diagrams in Figure 2.7. You can see that the valence of hydrogen is 1; oxygen, 2; nitrogen, 3; and carbon, 4. The situation is more complicated for phosphorus, in the third row of the periodic table, which can have a valence of 3 or 5 depending on the combination of single and double bonds it makes.

The molecules H_2 and O_2 are pure elements rather than compounds because a compound is a combination of two or more *different* elements. Water, with the molecular formula H_2O, is a compound. Two atoms of hydrogen are needed to satisfy the valence of one oxygen atom. **Figure 2.10c** shows the structure of a water molecule. (Water is so important to life that Chapter 3 is devoted to its structure and behavior.)

Methane, the main component of natural gas, is a compound with the molecular formula CH_4. It takes four hydrogen atoms, each with a valence of 1, to complete the valence shell of a carbon atom, with its valence of 4 **(Figure 2.10d)**. (We'll look at other carbon compounds in Chapter 4.)

Atoms in a molecule attract shared bonding electrons to varying degrees, depending on the element. The attraction of a particular atom for the electrons of a covalent bond is called its **electronegativity**. The more electronegative an atom is, the more strongly it pulls shared electrons toward itself. In a covalent bond between two atoms of the same element, the electrons are shared equally because the two atoms have the same electronegativity—the tug-of-war is at a standoff. Such a bond is called a **nonpolar covalent bond**. For example, the single bond of H_2 is nonpolar, as is the double bond of O_2. However, when an atom is bonded to a more electronegative atom, the electrons of the bond are not shared equally. This type of bond is called a **polar covalent bond**. Such bonds vary in their polarity, depending on the relative electronegativity of the two atoms. For example, the bonds between the oxygen and hydrogen atoms of a water molecule are quite polar **(Figure 2.11)**.

Figure 2.11

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Polar covalent bonds in a water molecule.
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Space-filling model of the water molecule, with a large oxygen atom bonded to two smaller hydrogens.

A space-filling model of a molecule of water, with the larger oxygen atom forming polar covalent bonds with each of two smaller hydrogen atoms. Shared electrons in the polar covalent bonds between oxygen and each hydrogen are pulled towards oxygen, giving oxygen a partial negative charge and each hydrogen atom a partial positive charge. Because oxygen is more electronegative than hydrogen, shared electrons are pulled towards oxygen.

Animation

Nonpolar and Polar Molecules

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Oxygen is one of the most electronegative elements, attracting shared electrons much more strongly than hydrogen does. In a covalent bond between oxygen and hydrogen, the electrons spend more time near the oxygen nucleus than near the hydrogen nucleus. Because electrons have a negative charge and are pulled toward oxygen in a water molecule, the oxygen atom has partial negative charges (indicated by the Greek letter δ with a minus sign, δ−, or “delta minus”), and the hydrogen atoms have partial positive charges (δ+, or “delta plus”). In contrast, the individual bonds of methane (CH4) are much less polar because the electronegativities of carbon and hydrogen are quite similar.

Ionic Bonds

In some cases, two atoms are so unequal in their attraction for valence electrons that the more electronegative atom strips an electron completely away from its partner. The two resulting oppositely charged atoms (or molecules) are called **ions**. A positively charged ion is called a **cation**, while a negatively charged ion is called an **anion**. (It may help you to think of the t in *cation* as a plus sign, and of *anion* as “a negative ion.”) Because of their opposite charges, cations and anions attract each other; this attraction is called an **ionic bond**. Note that the transfer of an electron is not, by itself, the formation of a bond; rather, it allows a bond to form because it results in two ions of opposite charge. Any two ions of opposite charge can form an ionic bond. The ions do not need to have acquired their charge by an electron transfer with each other.

This is what happens when an atom of sodium (N11a) encounters an atom of chlorine (C17l)**(Figure 2.12)**. A sodium atom has a total of 11 electrons, with its single valence electron in the third electron shell. A chlorine atom has a total of 17 electrons, with 7 electrons in its valence shell. When these two atoms meet, the lone valence electron of sodium is transferred to the chlorine atom, and both atoms end up with their valence shells complete. (Because sodium no longer has an electron in the third shell, the second shell is now the valence shell.) The electron transfer between the two atoms moves one unit of negative charge from sodium to chlorine. Sodium, now with 11 protons but only 10 electrons, has a net electrical charge of 1+; the sodium atom has become a cation. Conversely, the chlorine atom, having gained an extra electron, now has 17 protons and 18 electrons, giving it a net electrical charge of 1−; it has become a chloride ion—an anion.

Figure 2.12

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Electron transfer and ionic bonding.
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Electron transfer and ionic bonding.

The attraction between oppositely charged atoms, or ions, is an ionic bond. An ionic bond can form between any two oppositely charged ions, even if they have not been formed by transfer of an electron from one to the other.

The single valence electron of a sodium atom is transferred to join the 7 valence electrons in the third shell of a chlorine atom to form Sodium chloride (N a C l). The lone valence electron of a sodium atom with one electron in the third shell, eight electrons in the second shell, and two electrons in the first shell, is transferred to join the 7 valence electrons in the third shell of a chlorine atom to form Sodium chloride (N a C l). Each resulting ion has a completed valence shell. An ionic bond can form between the oppositely charged ions. N a plus has eight electrons in the second shell and two electrons in the first shell. C l minus has 8 electrons in the third shell, eight electrons in the second shell, and two electrons in the first shell.

Animation

Formation of Ions and Ionic Bonds

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Compounds formed by ionic bonds are called **ionic compounds**, or **salts**. We know the ionic compound sodium chloride (NaCl) as table salt **(Figure 2.13)**. Salts are often found in nature as crystals of various sizes and shapes. Each salt crystal is an aggregate of vast numbers of cations and anions bonded by their electrical attraction and arranged in a three-dimensional lattice. Unlike a covalent compound, which consists of molecules having a definite size and number of atoms, an ionic compound does not consist of molecules. The formula for an ionic compound, such as NaCl, indicates only the ratio of elements in a crystal of the salt. “NaCl” by itself is not a molecule.

Figure 2.13

A sodium chloride (NaCl) crystal.

Solid N a C l is a rigid, repeating structure with N a plus ions and C l minus ions.

The sodium ions (Na+) and chloride ions (Cl−) are held together by ionic bonds. The formula NaCl tells us that the ratio of Na+ to Cl− is 1:1.

Not all salts have equal numbers of cations and anions. For example, the ionic compound magnesium chloride (MgCl2) has two chloride ions for each magnesium ion. Magnesium (M12g) must lose 2 outer

electrons if the atom is to have a complete valence shell, so it has a tendency to become a cation with a net charge of 2+ (Mg^{2+}). One magnesium cation can therefore form ionic bonds with two chloride anions (Cl^-).

The term *ion* also applies to entire molecules that are electrically charged. In the salt ammonium chloride (NH_4Cl), for instance, the anion is a single chloride ion (Cl^-), but the cation is ammonium (NH_4^+), a nitrogen atom covalently bonded to four hydrogen atoms. The whole ammonium ion has an electrical charge of 1+ because it has given up 1 electron and thus is 1 electron short.

Environment affects the strength of ionic bonds. In a dry salt crystal, the bonds are so strong that it takes a hammer and chisel to break enough of them to crack the crystal in two. If the same salt crystal is dissolved in water, however, the ionic bonds are much weaker because each ion is partially shielded by its interactions with water molecules. Most drugs are manufactured as salts because they are quite stable when dry but can dissociate (come apart) easily in water. (In Concept 3.2, you will learn how water dissolves salts.)

Weak Chemical Interactions

In organisms, most of the strongest chemical bonds are covalent bonds, which link atoms to form a cell's molecules. But weaker interactions within and between molecules are also indispensable, contributing greatly to the emergent properties of life. Many large biological molecules are held in their functional form by weak interactions. In addition, when two molecules in the cell make contact, they may adhere temporarily by weak interactions. The reversibility of weak interactions can be an advantage: Two molecules can come together, affect one another in some way, and then separate.

Several types of weak chemical interactions are important in organisms. One is the ionic bond as it exists between ions dissociated in water, which we just discussed. Hydrogen bonds and van der Waals interactions are also crucial to life.

Hydrogen Bonds

Among weak chemical interactions, hydrogen bonds are so central to the chemistry of life that they deserve special attention. When a hydrogen atom is covalently bonded to an electronegative atom, the hydrogen atom has a partial positive charge that allows it to be attracted to a different electronegative atom with a partial negative charge nearby. This noncovalent attraction between a hydrogen and an electronegative atom is called a **hydrogen bond**. In living cells, the electronegative partners are usually oxygen or nitrogen atoms. **Figure 2.14** shows hydrogen bonding between water (H_2O) and ammonia (NH_3).

Figure 2.14

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A hydrogen bond.
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The hydrogen bond between water and ammonia.

The hydrogen bond between water (H_2O) and ammonia (NH_3) results from the attraction between the partial positive charge on the hydrogen atom of water and the partial negative charge on the nitrogen atom of ammonia. A space filling model of water is shown with a partial positive charge on the two hydrogen atoms and a partial negative charge on the oxygen atom. A space filling model of ammonia is also shown with a partial negative charge on the nitrogen atom and partial positive charges on the hydrogen atoms.

DRAW IT*Draw one water molecule hydrogen bonded to four other water molecules around it. Use simple outlines of space-filling models. Draw the partial charges on the water molecules and use dots for the hydrogen bonds.*

*For suggested answer, see*Appendix A*.*

Animation

Hydrogen Bonds

image

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Van der Waals Interactions

Even a molecule with nonpolar covalent bonds may have positively and negatively charged regions. Electrons are not always evenly distributed; at any instant, they may accumulate by chance in one part of a molecule or another. The results are ever-changing regions of positive and negative charge that enable all atoms and molecules to stick to one another. These **van der Waals interactions** are individually weak and occur only when atoms and molecules are very close together. When many such interactions occur simultaneously, however, they can be powerful: Van der Waals interactions allow the gecko lizard shown here to walk straight up a wall! The anatomy of the gecko's foot—including toes with hundreds of thousands of tiny hairs, each with multiple projections—maximizes surface contact with the wall. The van der Waals interactions between the foot molecules and the molecules of the wall's surface are so numerous that despite their individual weakness, together they can support the gecko's body weight.

A gecko lizard's foot.

Van der Waals interactions, hydrogen bonds, ionic bonds in water, and other weak interactions may form not only between molecules but also between parts of a large molecule, such as a protein or nucleic acid. The cumulative effect of weak interactions is to reinforce the three-dimensional shape of the molecule. (You will learn more about the very important biological roles of weak interactions in Figures 5.18 and 5.24.)

Molecular Shape and Function

A molecule has a characteristic size and shape, which are key to its function in the living cell. A molecule consisting of two atoms, such as H2 or O2, is always linear, but most molecules with more than two atoms have more complicated shapes. These shapes are determined by the positions of the atoms' orbitals **(Figure 2.15)**. When an atom forms covalent bonds, the orbitals in its valence shell undergo rearrangement.

Figure 2.15

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Molecular shapes due to hybrid orbitals.
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Molecular shapes due to hybrid orbitals.

- Part a. Hybridization of orbitals. The single s and three p orbitals of a valence shell involved in covalent bonding combine to form four teardrop-shaped hybrid orbitals. These orbitals extend to the four corners of an imaginary tetrahedron.
- Part b. A space-filling model of a water molecule shows a large oxygen atom covalently bonded to two smaller white hydrogen atoms. A ball and stick model of water indicates that the V-shaped angle between the hydrogen atoms in the water molecule is 104.5 degrees. A Hybrid-Orbital model of water shows that two of the hybrid orbitals are shared with hydrogens in polar covalent bonds, leaving one pair of unbounded electrons in oxygen's outer shell.
- Part c. A space-filling model of methane shows a central carbon atom covalently bonded to four hydrogen atoms. A ball-and-stick model represents the shape of the methane molecule as a tetrahedron with a central carbon and hydrogen atoms at all four corners of the tetrahedron. A Hybrid-Orbital model indicates all four hybrid orbitals on the carbon atom are shared with hydrogen atoms.

For atoms with valence electrons in both s and p orbitals (review Figure 2.8), the single s and three p orbitals form four new hybrid orbitals shaped like identical teardrops extending from the region of the atomic nucleus, as shown in Figure 2.15a. If we connect the larger ends of the teardrops with lines, we have the outline of a geometric shape called a tetrahedron, a pyramid with a triangular base.

For water molecules (H_2O), two of the hybrid orbitals in the oxygen's valence shell are shared with hydrogens. The other two hybrid orbitals are occupied by lone (unbonded) pairs of electrons (see Figure 2.15b). The result is a molecule shaped roughly like a V, with its two covalent bonds at an angle of 104.5°.

The methane molecule (CH_4) has the shape of a completed tetrahedron because all four hybrid orbitals of the carbon atom are shared with hydrogen atoms (see Figure 2.15b). The carbon nucleus is at the center, with its four covalent bonds radiating to hydrogen nuclei at the corners of the tetrahedron. Larger molecules containing multiple carbon atoms, including many of the molecules that make up living matter, have more complex overall shapes. However, the tetrahedral shape of a carbon atom bonded to four other atoms is often a repeating motif within such molecules.

Molecular shape is crucial: It determines how biological molecules recognize and respond to one another with specificity. Biological molecules often bind temporarily to each other by forming weak interactions, but only if their shapes are complementary. Consider the effects of opiates, drugs such as morphine and heroin derived from opium. Opiates relieve pain and alter mood by weakly binding to specific receptor molecules on the surfaces of brain cells. Why would brain cells carry receptors for opiates, compounds that are not *endogenous*, made by the body? In 1975, this question was answered by the discovery of endorphins (or "endogenous morphines"). Endorphins are signaling molecules made by the pituitary gland that bind to the receptors, relieving pain and producing euphoria during times of stress, such as intense exercise. Opiates have shapes similar to endorphins and can bind to endorphin receptors in the brain. That is why opiates and endorphins have similar effects **(Figure 2.16)**. The role of molecular shape in brain chemistry illustrates how biological organization leads to a match between structure and function, one of biology's unifying themes.

Figure 2.16

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A molecular mimic.
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Two figures illustrate morphine mimicking endorphin.

Morphine affects pain perception and emotional state by mimicking the brain's natural endorphins.

Parts are as follows.

- Figure a compares the structures of endorphin and morphine. A key part of the endorphin molecule binds to receptor molecules on target cells in the brain. Part of the morphine molecule is a close match. These both largely involve carbon, hydrogen, and oxygen.
- Figure b is of binding to endorphin receptors. Both endorphin and morphine can bind to endorphin receptors on the surface of a brain cell. Here, the endorphin receptors are in the lipid

bilayer of the cell, with 1 end inside the cell and the other end available to bind to natural endorphin or morphine.

Interview

Interview with Candace Pert: Discovering opiate receptors in the brain

A photo of Candace Pert.